



Mechanical Engineering Design Project

Final Report

Course: ME 3202

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Group Number: B6

Project Title: Design a Belt Drive for the Starter Motor of a Car Engine

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Chapter 1

Clarify Problem Specifications and Constraints

The objective of this project is to design a belt drive system that efficiently transmits power from the engine to the starter motor of a car engine. The design aims to ensure durability, minimal maintenance and optimal performance under various operating conditions.

1.0.1 Design requirements

1. Strength and durability to withstand high torque and engine conditions.
2. Correct tensioning for optimal power transfer.
3. Resistance to environmental factors like temperature, moisture, and oils.
4. Good grip and alignment to prevent slipping and premature wear.
5. Compact and space-efficient design to fit within the engine bay.
6. Low maintenance need with periodic replacement and inspection.

1.0.2 Design specifications

1. Power transmission: From starter motor to engine 1KW to 3KW.
2. Noise Level: Up to 75 dB under full load.
3. Belt Width: 5 mm to 30 mm.
4. Belt Length: Dependent on engine layout and center distance. Commonly 600 to 1200 mm
5. Pulley Diameter (Driver): 50 mm to 100 mm
6. Pulley Diameter (Driven): 60 mm to 120 mm
7. Tensile Strength: >2000 Nm for durability
8. Torque: Starter motor torque 10 to 50 Nm.

9. Speed Ratio: 1:2 (depending on the starter motor design)
10. Operating Temperature: -40°C to 120°C (up to 150°C in some cases)
11. Service Life: 50,000 to 100,000 miles (80,000 to 160,000 km) or 5 years

1.0.3 Engineering Constraints

1. Strength & Durability
2. Performance Criteria
3. Manufacturing Feasibility
4. Lifecycle Considerations
5. Material Selection
6. Belt Tensioning & Slippage Control
7. Noise & Vibration Control
8. Maintenance & Accessibility
9. Environmental Resistance
10. Cost-Effectiveness

Chapter 2

Research and Investigate

2.0.1 Literature Review

➤ **Belt-Driven System:**

The system uses two belts. One belt connects the crankshaft pulley of the internal combustion engine (ICE) to the MGU, transferring torque to start or assist the engine. The second belt transmits power to vehicle accessories.

➤ **Motor-Generator Unit (MGU):**

The MGU operates as an electric motor when starting the engine and can also act as a generator when the engine is running, recharging the vehicle's battery. The MGU drives the crankshaft and various vehicle accessories, such as the air conditioning compressor, water pump, hydraulic power steering pump, and transmission pumps.

➤ **Compact and Efficient Design:**

The system design is intended to be compact and cost-effective, avoiding the need for multiple belts or complex tensioner systems found in conventional systems.

➤ **System Design and Requirements:**

- Target: Start engine at -20°C, generate 3-4 kW at medium speeds, and 1 kW at 6000 rpm.
- Crank at least at 400 rpm at room temperature.
- 42 V output for starter-generator system.[12]

➤ **Motoring Torque Requirements:**

The BAS system enables a quick and quiet engine start and requires significant peak motoring torque to ensure rapid engine startup, particularly under cold conditions. The BAS system enables a quick and quiet engine start and requires significant peak motoring torque to ensure rapid engine startup, particularly under cold conditions.

➤ **Mechanical Drive:**

-Drive System: Poly-V belt with eight ribs chosen.

-Belt drive chosen due to packaging flexibility and minimal engine modifications. Allows larger motor packaging near the alternator.

2.0.2 Theoretical Background

The application of mechanical engineering concepts, such as power transmission, torque analysis, pulley diameter, angle of contact, belt tension, and material selection, is necessary when designing a belt drive system for an automobile's starting motor. The main ideas and governing equations are listed here.

$$\text{Angle of contact } \theta_d = \pi - 2 \sin^{-1} \frac{D-d}{2C}$$

$$\text{Belt Lenth } L = [4C^2 - (D - d)^2]^{\frac{1}{2}} + \frac{1}{2}(D\theta_D + d\theta_d)$$

Here,

- Diameter of large pulley, $D = 120\text{mm}$
- Diameter of small pulley, $d = 80\text{mm}$
- Center distance, $C = 200$
- Angle of contact, θ
- $L =$ Belt length

$$\text{Speed Ratio } \frac{N_2}{N_1} = \frac{d_1+t}{d_2+t} \left(1 - \frac{s}{100}\right)$$

$$\text{Centrifugal tension in the belt: } T_c = m \cdot v^2$$

Power transmitted by a belt: $P = (T_1 - T_2)v$

- d_1 = Diameter of the driver
- d_2 = Diameter of the Driven
- N_1 = Speed of the driver in rpm
- N_2 = Speed of the driven in rpm
- t = Thickness
- s = Total slip (5% each pulley)

Chapter 3

Generate Alternative Designs

We explored three different design concepts for the belt drive system, each with its own feasibility and advantages.

3.0.1 Concept 1: V-Belt Drive

V-belts are trapezoidal belts that provide high torque transmission with minimal slip due to their wedging action in pulley grooves. They provide the best efficiency, compact design, and good shock absorption.

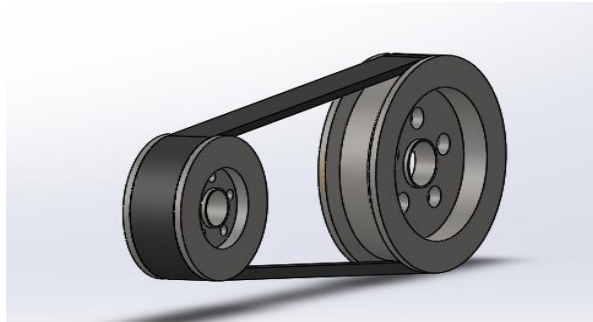


Fig: V-Belt Drive

3.0.2 Concept 2: Timing Belt

A timing belt can also be used here. It has minimal slip due to its geometry. But its cost high.

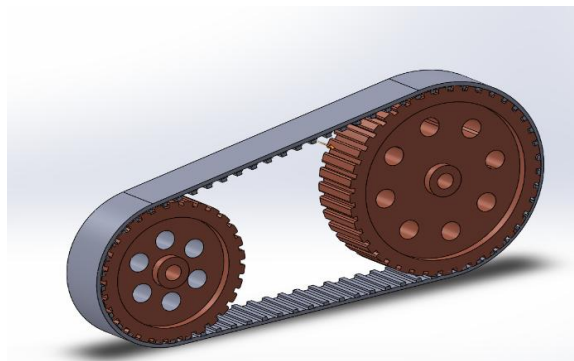


Fig: Timing Belt

3.0.3 Concept 3: Flat Belt

It has the similar specification with V-Belt but squared shape. In a long drive system, a flat belt is used in industries.

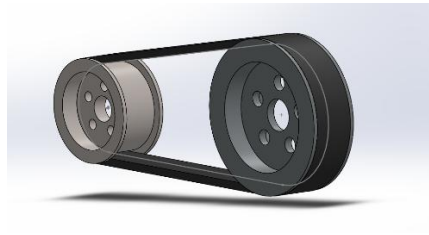


Fig. Flat Belt

3.0.4 Comparison of Alternatives

Design	Slipping	Length, L	$\sigma_{\text{allowable}}$	Power Transmission	Cost per unit length
V-Belt	Less Slippage	716.15mm	13.16MPa	11.43 KW	Moderate
Timing belt	Good Grip	731mm	8.33MPa	7.6 KW	High
Flat Belt	More Slippage	716.15mm	8.33MPa	5 KW	Low

From an overall comparison, it can be observed that the V-belt is ideal for this design.

Chapter 4

Choose and Justify the Optimal Solution

4.0.1 Decision Criteria

A key factor for this design is power transmission at very high speed. This deformation and stress will occur in the belt. The Material property plays a vital role in it. Slip is also a major concern. The design should be cost-effective when it comes to mass production.

4.0.2 Final Design Selection

V-Belt drive is taken for the final design. V-belts have a better power transmission ratio, less slippage, good availability on the market, and good reliability over other designs.

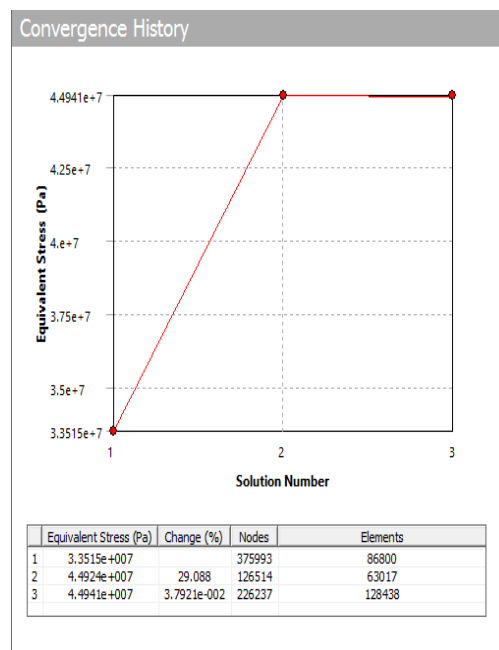
4.0.3 Analytical Calculations

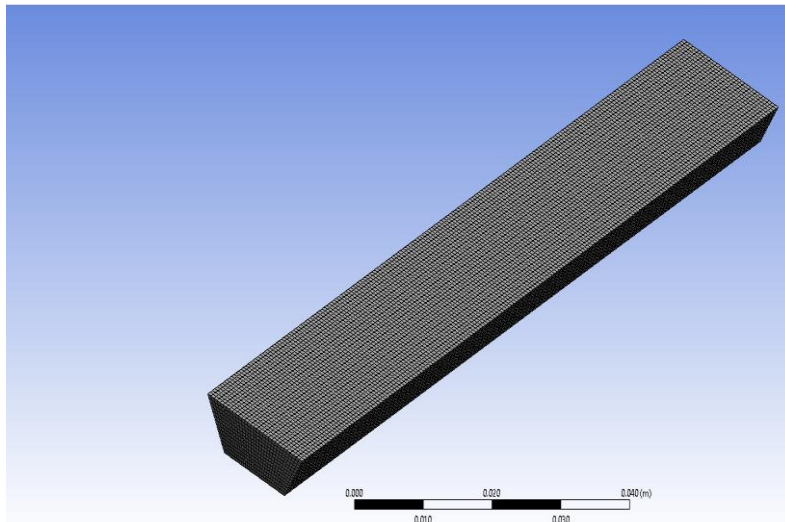
Include key design equations, stress calculations, and factor of safety estimations.

4.0.4 FEA Analysis and Verifications

Mesh Sensitivity Analysis

Here is the analysis of the mesh quality and convergence behavior for stress and deformation results.





Validation with Experimental Results

As we did not find any relevant papers or patented designs for our V-belt design, we could not validate our design.

Error Analysis

Theoretical value	Numerical Value	Error (%)
1.55	1.284	13.49%

We found around 13.5% error between our theoretical design value and the numerical value found by conducting FEA analysis.

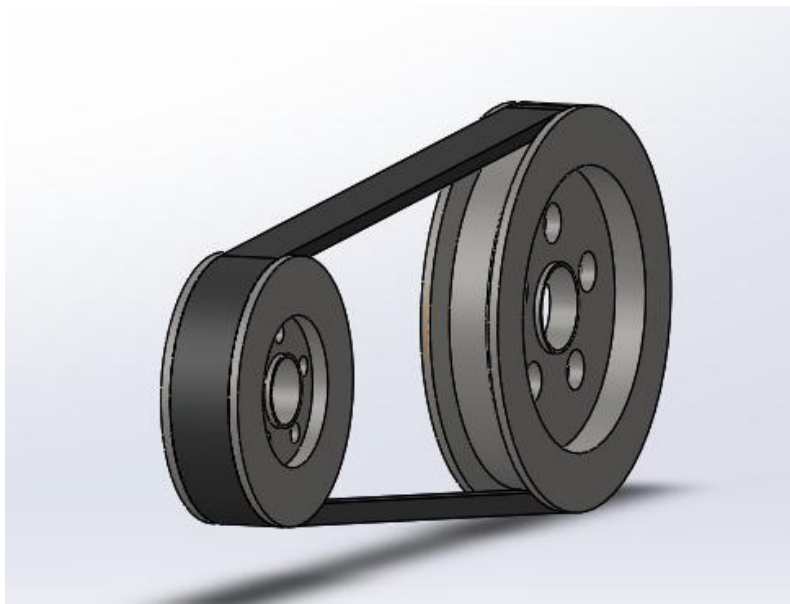
Chapter 5

Develop a Prototype

We created the CAD model of the final gear train design. Here are the detailed drawings with dimensions and technical specifications.

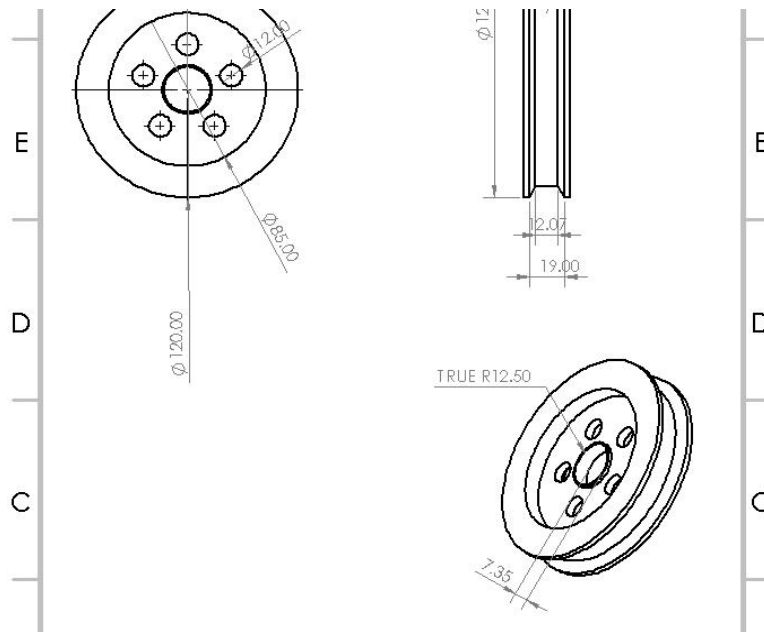
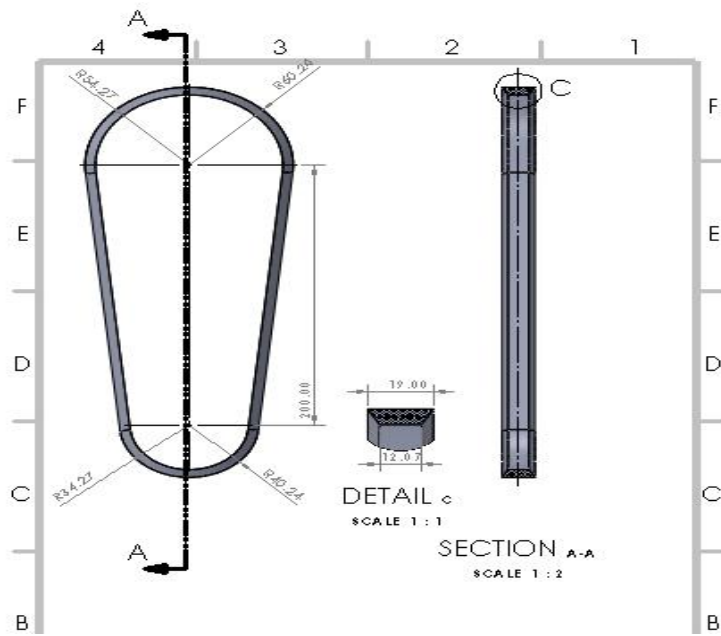
5.0.1 Technical Specifications

Here is the final CAD model of our v-Belt design.



5.0.1 Technical Drawings

Here is the 2D isometric view of our final design, including all relevant dimensions. This drawing showcases the key components of the belt drive system, including the pulleys, shaft, and belt alignment. The dimensions have been carefully calculated to ensure optimal performance and fit within the required specifications.



Chapter 6

Test and Evaluate

We conducted a theoretical analysis and performed simulations using ANSYS FEA software, where we applied varying loads to assess the performance and durability of the belt drive system. The simulation focused on stress analysis, torque calculations, gear ratio verification, and efficiency estimation.

During the simulation, we varied the load and observed the stress distribution and failure points in the system. The results showed that the design began to fail under a load of 1200 N, which we confirmed by monitoring stress and deformation. As a result, we decided to stop the test at 1300 N to avoid any damage to the system. Below is the table that shows the load change and corresponding stress levels during the simulation.

Force/load (N)	Deformation (m)	FOS	Equivalent Stress (MPa)
700	0.000267	1.834	26.2
800	0.0031	1.61	29.9
900	0.000344	1.4268	33.7
1000	0.000382	1.284	37.45
1100	0.00042	1.1674	41.2

1200	0.00045	1.07	44.9
1300	0.0004967	0.9872	48.68

Chapter 7

Redesign the Solution

Based on the test results, several modifications can improve the performance of the belt drive system. Identified weaknesses include belt slippage, excessive wear, noise, power loss, and overheating. To address these, an automatic tensioning mechanism can prevent belt slippage, while using more durable materials like Kevlar or hardened steel for the belt and pulleys will reduce wear. Precision alignment and optimized pulley sizes can minimize noise, vibration, and power loss. Adding a cooling system can help reduce overheating, and regular maintenance checks will ensure long-term efficiency. These refinements will enhance reliability, efficiency, and overall system performance.

7.0.1 Identified Weaknesses

Highlight You'll focus on any shortcomings or issues that were discovered during the testing and validation phase of your belt drive design. It's important to explain the specific areas where the design did not perform as expected or where there was room for improvement. Here's how you can approach it:

1. **Belt Slip:** Occurs under high torque, reducing power transmission.
2. **Excessive Wear:** Fast wear of the belt and pulleys due to improper materials or high temperatures.
3. **Noise and Vibration:** Caused by misalignment or improper tension.
4. **Power Loss:** Due to friction, improper pulley sizes, or material inefficiencies.
5. **Overheating:** Caused by high friction or excessive load.

7.0.2 Proposed Improvements

Suggest you propose modifications to address the weaknesses identified above and enhance the performance and reliability of the system. You can suggest both design changes and operational adjustments.

1. **Better Tensioning:** Implement automatic/manual tensioning to prevent slippage.
2. **Durable Materials:** Use higher-grade materials for the belt and pulleys (e.g., Kevlar, hardened steel).
3. **Precision Alignment:** Ensure better alignment of pulleys and shafts to reduce noise and vibration.

4. **Optimized Pulley Size:** Adjust pulley size and gear ratios to minimize power loss.
5. **Cooling System:** Add air or oil cooling to prevent overheating.
6. **Regular Maintenance:** Implement a maintenance schedule to keep the system efficient.
7. **Variable Load Testing:** Test under different load conditions to identify further optimizations.

Chapter 8

Communicate Your Achievements

The designed belt drive system ensures efficient power transmission for the starter motor, achieving the required speed ratio and torque capacity while minimizing slippage. With durable, heat-resistant materials, the system operates reliably under high temperatures and harsh conditions. Its adjustable tensioning mechanism enhances stability, preventing performance losses over time. The final design provides a cost-effective and low-maintenance solution with minimal noise and vibration, improving engine startup efficiency. This project contributes to automotive engineering by optimizing starter motor performance and can be further improved with advanced materials, automated tensioning, and noise-reduction techniques. Future adaptations may also extend its application to electric and hybrid vehicles.

8.0.1 Summary of Results

The designed belt drive system for the automobile starter motor successfully meets the required performance, durability, and efficiency standards. The final design ensures optimal power transmission efficiency, proper speed ratio, and effective tensioning to prevent slippage. The system components, including the belt, pulleys, and tensioning mechanism, were carefully selected to withstand high torque loads and harsh environmental conditions while maintaining low noise and vibration levels. The material selection and design specifications balance strength, cost-effectiveness, and longevity, ensuring a reliable system with minimal maintenance requirements.

8.0.2 Impact and Applications

This project contributes to the automotive industry by offering a cost-effective and efficient solution for starter motor power transmission. The designed system enhances engine startup reliability while reducing energy losses, improving overall vehicle performance. The findings of this project can be applied to automotive belt drive systems, ensuring better durability and operational efficiency. Additionally, the project serves as a foundation for future research in optimizing belt drive mechanisms for hybrid and electric vehicle applications.

8.0.3 Future Work

Several improvements can be made to further enhance the performance and reliability of the design:

1. **Advanced Material Research:** Exploring high-performance composite materials to improve durability and heat resistance.
2. **Dynamic Load Analysis:** Conducting real-time testing under varying operational conditions to optimize belt tensioning and performance.
3. **Noise Reduction Techniques:** Implementing vibration-damping technologies to further reduce noise levels.

4. Automated Tensioning System: Developing a self-adjusting tensioner for better longevity and efficiency.
5. Application in EVs: Adapting the design for use in electric vehicle starter systems, integrating lightweight materials and high-efficiency transmission mechanisms.

These enhancements will ensure greater efficiency, longer service life, and better adaptability for future automotive technologies.

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